

Forecasting Brief — Retail Store Sales

What the data shows

The store's daily sales show a strong long-term upward trend from 2021 through 2024. There is also a clear weekly pattern, which is why using the same day's sales from the previous week produces a strong forecast. Short-term daily variation remains around these patterns.

The time-series analysis also shows that the raw sales series is non-stationary, while first differencing, seasonal differencing and the STL residual are stationary according to the ADF test.

Model performance

Three simple baselines were tested before training the model:

- Overall mean: MAE = 522.99
- Yesterday's sales: MAE = 195.79
- Same day last week: MAE = 180.50

The Random Forest achieved:

- Random KFold MAE = 138.22
- TimeSeriesSplit MAE = 188.01

The random KFold result looks better than the seasonal-naive baseline, but it is not a reliable estimate for future forecasting because random splitting mixes observations from different time periods.

The TimeSeriesSplit result is the more realistic evaluation. Its MAE of approximately 187.22 is worse than the seasonal-naive baseline of approximately 180.50.

Recommendation

Do not deploy the Random Forest yet.

For the current dataset, I would deploy the seasonal-naive forecast instead. It is simple, easy to explain, respects the weekly sales pattern, and performs slightly better than the Random Forest under the time-aware evaluation.

What could improve the forecast

The most useful improvement would likely come from additional information that explains changes in sales beyond their historical pattern, such as holidays, special events, major promotions with measurable effects, stock availability, pricing, or other store-specific events.

The current temperature variable does not provide strong independent signal after weekly seasonality is removed, and the promo flag shows very little difference in average sales.

Limitation

The Random Forest evaluation is based on historical cross-validation rather than a live future forecasting environment. In addition, the dataset contains only four years of observations for one store, so unusual future events may not be represented in the training data.

Final decision

Use the seasonal-naive baseline for now and continue investigating additional external information before replacing it with the Random Forest model.